VLM benchmark: 200-image pilot vs full test set

MRIModelBench · qwen2.5vl:3b run locally through Ollama · notebooks/ollama_vlm.ipynb

> **In short:** we first tested the vision-language model on a 200-image sample, then on all 1,584 test images. The full run confirmed the pilot. Both accuracies stayed inside the pilot's margin of error, and every pattern the pilot found was still there. The pilot slightly overestimated zero-shot accuracy, mostly because its 200 images happened to be a little easier than average. The full-test-set numbers are the ones to report.

1. What the two runs were

Both runs used the same model (qwen2.5vl:3b), the same prompt and the same settings (temperature 0, seed 0). Each run had two setups. **Zero-shot** shows the model only the scan and a description of the four classes. **Few-shot** also shows 8 labelled example scans from the training set, 2 per class. The model is never trained: its weights are not changed in either run.

	200-image pilot	Full test set
Images	200: a random sample of 50 per class, drawn with a fixed seed	All 1,584 test images (386 glioma, 398 meningioma, 400 notumor, 400 pituitary)
Purpose	Check that the setup works, estimate the run time, get a first result	The final result, on the same test set as the naive baseline and the decision tree
Run time (4 GB laptop GPU)	About 25 minutes for both setups	About 3 hours 20 minutes (82 min zero-shot, 116 min few-shot)
Margin of error on accuracy*	About ± 7 percentage points	About ± 2.5 percentage points

The 200 pilot images are part of the 1,584, so the full run classified them a second time. That overlap lets us check two separate things: whether the pilot's *images* were representative (section 4), and whether the model gives the same *answers* when run again (section 5).

* The ± of a 95% confidence interval, $1.96 \times \sqrt{p(1-p)/n}$. It shrinks with more images: 200 images give roughly ± 7 points, 1,584 roughly ± 2.5.

2. Headline results

Setup	Run	Images	Accuracy (± 95%)	Macro F1	Tumour scans called healthy
Zero-shot	200-image pilot	200	0.470 ± 0.069	0.408	12 / 150 (8.0%)
	Full test set	1,584	0.426 ± 0.024	0.371	146 / 1,184 (12.3%)
Few-shot	200-image pilot	200	0.395 ± 0.068	0.352	70 / 150 (46.7%)
	Full test set	1,584	0.396 ± 0.024	0.341	641 / 1,184 (54.1%)

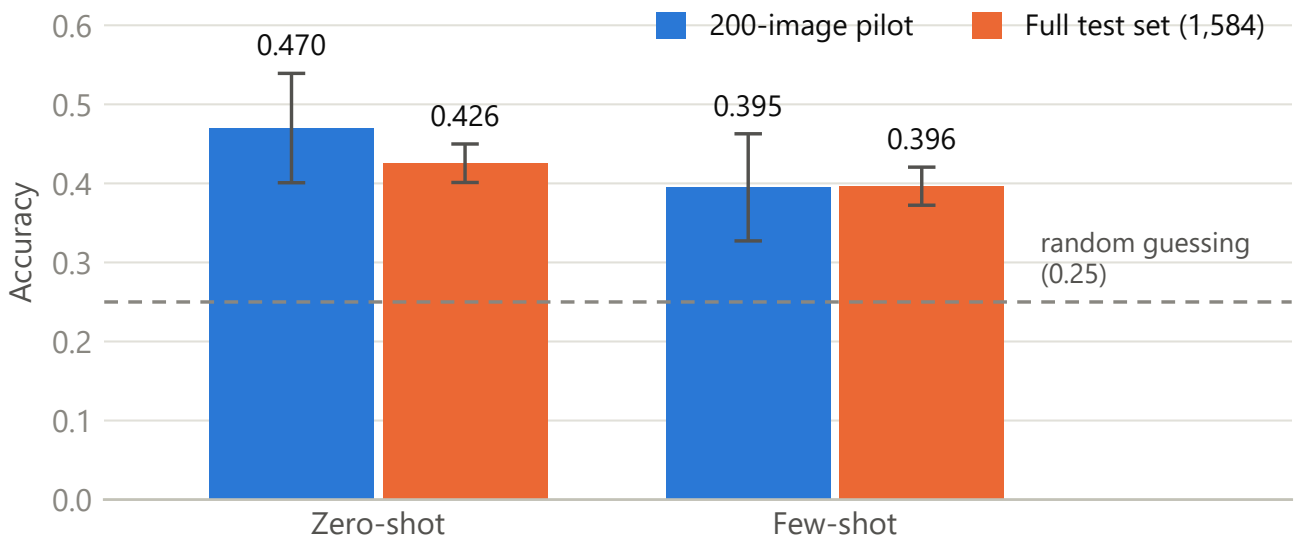


Figure 1. Accuracy of each run with its 95% margin of error (whiskers). The full-run bar falls inside the pilot's whisker for both setups. The dashed line is a random guess between the four classes.

- **Zero-shot** fell from 0.470 to 0.426 (−4.4 points), within the pilot's ± 6.9. Macro F1 fell from 0.408 to 0.371.
- **Few-shot** was almost unchanged: 0.395 → 0.396 accuracy and 0.352 → 0.341 macro F1.
- **The ranking held.** Zero-shot beats few-shot in both runs, both beat random guessing (0.25), and both stay far below the decision tree (0.77 accuracy).
- **Tumours called healthy**, the most serious error, rose a little in both setups: zero-shot from 8.0% to 12.3%, few-shot from 46.7% to 54.1%. With only 150 tumour scans in the pilot, both changes are within its margin of error. The zero-shot change sits right at its edge.

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3. Did the patterns hold?

The pilot's main findings were about *how* the model goes wrong, not only how often. All of them held on the full test set.

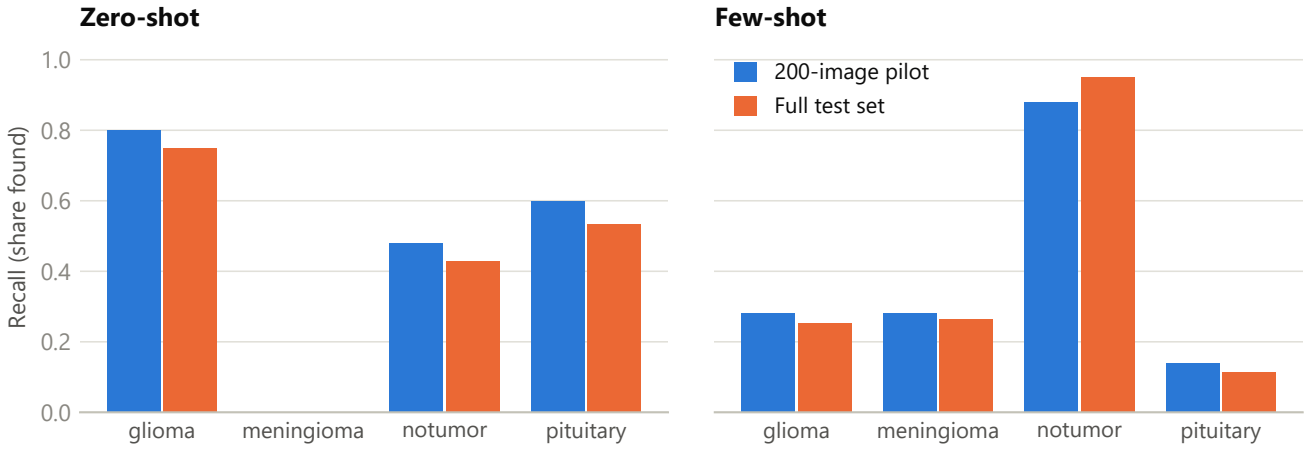


Figure 2. Recall per true class: the share of each class's scans the model labelled correctly. The shape of every panel is the same in both runs. Exact values are in the table below.

	Zero-shot recall		Few-shot recall		Zero-shot: share of answers		Few-shot: share of answers	
Class	Pilot	Full	Pilot	Full	Pilot	Full	Pilot	Full
glioma	0.80	0.75	0.28	0.25	55%	57%	18%	16%
meningioma	0.00	0.00	0.28	0.26	0%	0%	18%	15%
notumor	0.48	0.43	0.88	0.95	18%	20%	57%	64%
pituitary	0.60	0.53	0.14	0.11	28%	23%	8%	5%

"Share of answers" is how often the model gave each label. Each class is about 25% of the test set, so a share far above 25% shows a bias towards that label.

- **Zero-shot never says "meningioma".** It gave that answer 0 times in the pilot and 0 times in 1,584 images, so meningioma recall is 0.00 in both.
- **Zero-shot leans towards "glioma".** It answered glioma 55% of the time in the pilot and 57% in the full run, against a true share of about 25%.
- **Few-shot leans towards "notumor".** Its share of notumor answers rose from 57% to 64%, so it finds almost every healthy scan (recall 0.95) by calling most scans healthy.
- **Few-shot misses pituitary tumours.** Pituitary recall was 0.14 in the pilot and 0.11 in the full run, against 0.53 for zero-shot.

4. Why zero-shot accuracy dropped

A gap between the pilot and the full run can have two causes. Either the 200 sampled images were easier or harder than the test set as a whole (*sampling luck*), or the model answered the same images differently the second time (*run-to-run noise*). Because the full run includes the 200 pilot images, we can separate the two:

Setup	Pilot run (200 images)	Full run, same 200 images	Full run, other 1,384 images	Full run, all 1,584
Zero-shot	0.470	0.460	0.421	0.426
Few-shot	0.395	0.390	0.397	0.396

For zero-shot, the same 200 images scored 0.460 in the full run. So only 1.0 of the 4.4-point drop is run-to-run noise. The remaining 3.4 points are sampling luck: the 1,384 images outside the pilot scored 0.421, so the pilot drew slightly easier images than average. For few-shot, all four numbers are within one point of each other.

5. Is the model reproducible?

With temperature 0 and a fixed seed, the model should in principle give the same answer every time. In practice a small share of answers changed between the two runs:

Setup	Same answer	Answer changed	Right → wrong	Wrong → right	Wrong → other wrong
Zero-shot	92.0%	16 of 200	7	5	4
Few-shot	94.0%	12 of 200	4	3	5

The changes almost cancel out (zero-shot: 7 answers became wrong and 5 became right), which is why the overall scores barely moved. The most likely cause is that the model runs partly on the CPU and partly on the GPU. Small differences in floating-point rounding between runs can flip scans where two labels were nearly tied. As a result, individual predictions are not fully reproducible, but the scores and biases are.

6. What this means for the report

- **Use the full-test-set numbers**: zero-shot accuracy 0.426, macro F1 0.371; few-shot accuracy 0.396, macro F1 0.341. They have a third of the pilot's uncertainty and use the same 1,584 images as the decision tree (0.77), so the comparison is like for like.
- **The pilot was a fair preview.** Its conclusions carry over unchanged: the VLM beats random guessing but is far behind a trained model, zero-shot is biased towards glioma and never predicts meningioma, and few-shot examples make the model worse by pushing it towards "notumor".
- **A pilot is a sound way to work.** About 25 minutes on 200 images predicted the 3 hour 20 minute full result to within its stated margin of error. For zero-shot, the pilot estimate was 4 points too optimistic, so treat pilot numbers as estimates rather than final results.